9.2 Amplituden- und Phasenverläufe

Ziel: Bodediagramm : $\boxed{a_{dB} = 20 \log_{10} |\underline{A}|}$

$\varphi = \varphi_Z - \varphi_N$

9.2.1 Konstante K

$\underline{A} = K$ z. B $\underline{A} = 5$

$a_{dB} = 20 \cdot \log_{10} |K| = \ldots$

$\varphi = 0°$

9.2.2 Negative Konstante

$\underline{A} = -K$, z.B $\underline{A} = -5$

$a_{dB} = 20 \cdot \log_{10} |K| =$

$\varphi = \pm 180°$

9.2.3 Differentiator $\underline{A} = j\omega T$

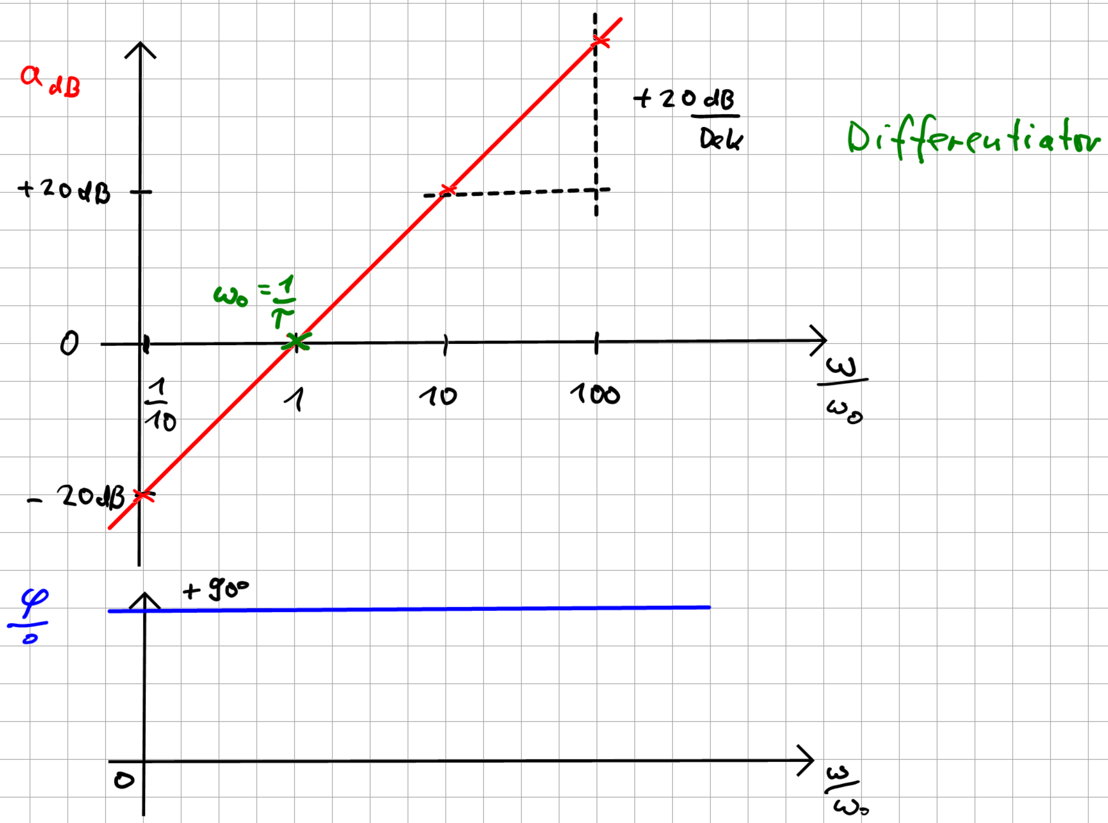
$a_{dB} = 20 \cdot \log_{10} |j\omega T|$

$= 20 \cdot \log_{10} (\omega \underset{1s}{T})$ T ist Zeitkonstante

Bsp: $T = 1s$

$\omega_0 = \frac{1}{T} = \frac{1}{1s} = 1 \cdot \frac{1}{s}$

$\frac{\omega}{s}$	$\frac{1}{10}$	1	10	100
a_{dB}	$-20\,dB$	$0\,dB$	$+20\,dB$	$+40\,dB$



9.2.4 Integrator $\underline{A} = \frac{1}{j\omega T}$

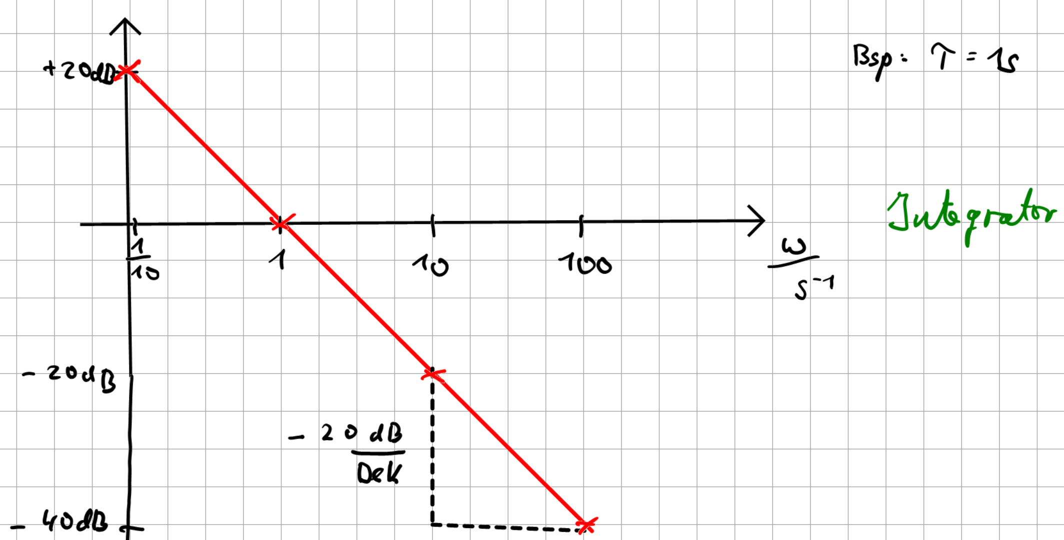
$$a_{dB} = 20 \cdot \log_{10} \left| \frac{1}{j\omega T} \right| \qquad T = \frac{1}{\omega_0}$$

$$\log \frac{a}{b} = \log a - \log b$$

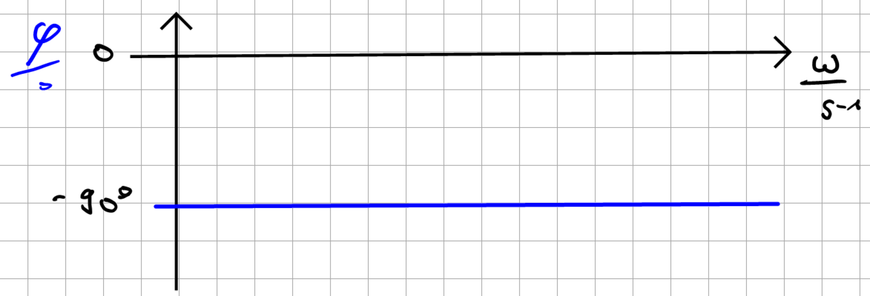
$$= 20 \cdot \log_{10} |1| - 20 \log |j\omega T|$$

$$= 0 - 20 \cdot \log(\omega T)$$

wie Differentiator, aber $\cdot(-1)$



$$\varphi_{\underline{A}} = \sphericalangle\left(\frac{1}{j\omega T}\right) = \varphi_Z - \varphi_N = 0° - 90° = -90°$$



9.2.7 $\underline{A} = 1 + j\omega T$ („Z1")

$$a_{dB} = 20 \cdot \log_{10} |1 + j\omega T|$$

$T = \frac{1}{\omega_0}$ ω_0 = Eckfrequenz

$$= 20 \log_{10} \left|1 + j\frac{\omega}{\omega_0}\right|$$

$$= 20 \log_{10} \sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2}$$

$$\varphi_A = \sphericalangle(1 + j\omega T)$$

Fallunterscheidungen

1. Fall: $\omega = \omega_0$

$$a_{dB} = 20 \cdot \log \sqrt{1 + \frac{\omega_0}{\omega_0}} = 20 \cdot \log_{10} \sqrt{1+1}$$

$$= 20 \cdot \log_{10} \sqrt{2}$$

$$= +3\ dB$$

$$\varphi_A = \sphericalangle\left(1 + j\frac{\omega_0}{\omega_0}\right) = \sphericalangle(1 + j1)$$

$$= +45°$$

2. Fall: $\omega \ll \omega_0$

$$a_{dB} = 20 \cdot \log \sqrt{1 + \cancel{\left(\frac{\omega}{\omega_0}\right)}^2} = 20 \log_{10} 1 = 0 \, dB$$

$$\varphi_A = \sphericalangle \left(1 + j \cancel{\frac{\omega}{\omega_0}}\right) = 0°$$

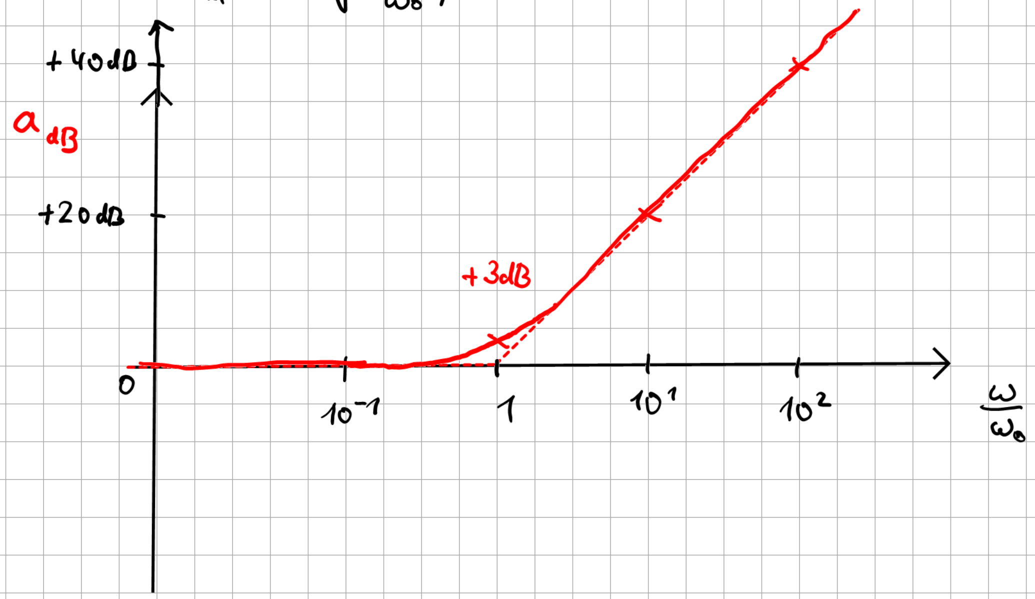
3. Fall: $\omega \gg \omega_0$

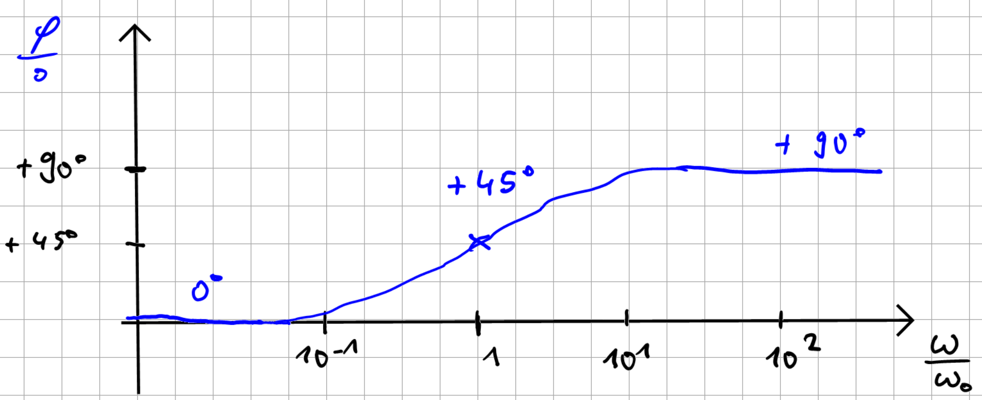
$$a_{dB} = 20 \cdot \log_{10} \sqrt{\cancel{1+} \left(\frac{\omega}{\omega_0}\right)^2}$$

$$= 20 \cdot \log_{10} \left(\frac{\omega}{\omega_0}\right) = 20 \cdot \log_{10} (\omega \cdot T)$$ wie Differentiator

$\frac{\omega}{\omega_0}$	1	10	100
a_{dB}	0	+20 dB	+40 dB

$$\varphi_A \left(1 + j \frac{\omega}{\omega_0}\right) = +90°$$



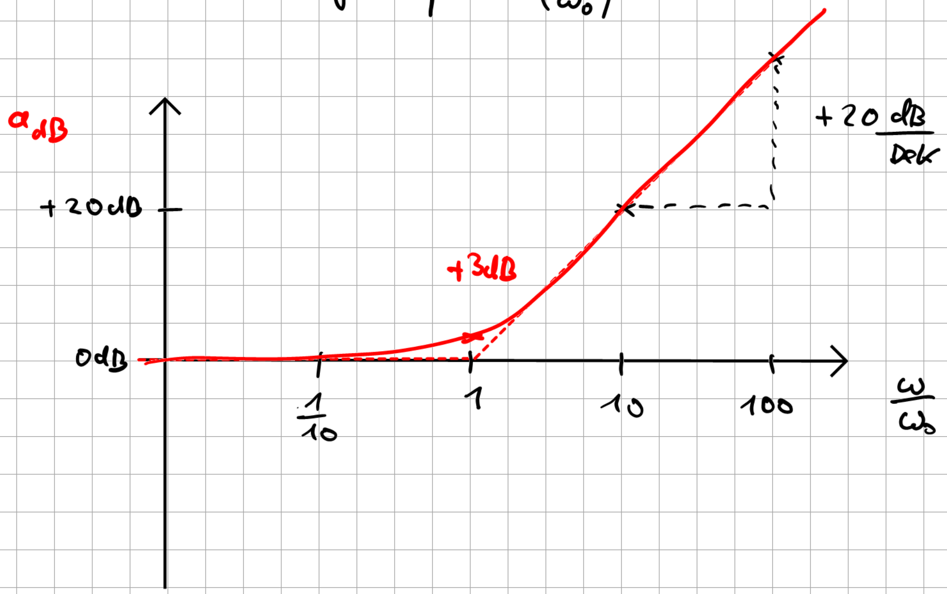


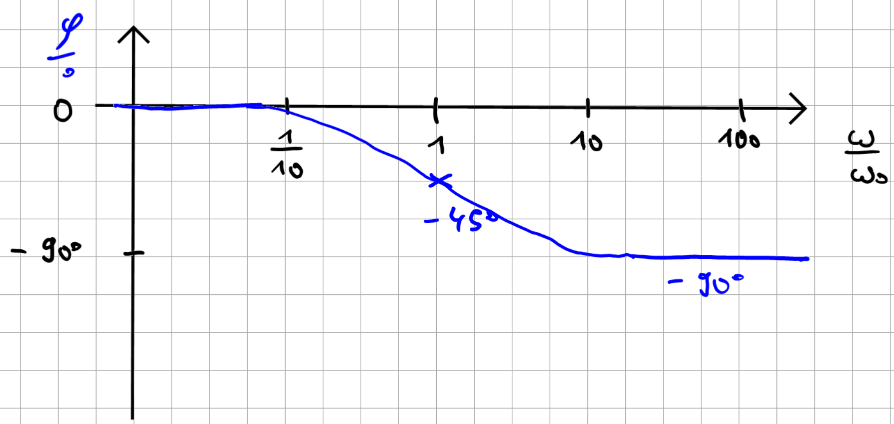
9.2.8 Funktion: $\underline{A} = 1 - j\omega T$ („Z2")

$$a_{dB} = 20 \cdot \log_{10} |1 - j\omega T| = \qquad T = \frac{1}{\omega_0}$$

$$= 20 \cdot \log_{10} \left|1 - j\frac{\omega}{\omega_0}\right|$$

$$= 20 \log_{10} \sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2}$$ wie bei „Z1"





9.2.9 Allpass $\underline{A} = \frac{1 - j\omega T}{1 + j\omega T}$

$$a_{dB} = 20 \cdot \log \left| \frac{1 - j\omega T}{1 + j\omega T} \right| = \qquad T = \frac{1}{\omega_0}$$

$$= 20 \cdot \log_{10} |1 - j\omega T| - 20 \cdot \log_{10} |1 + j\omega T|$$

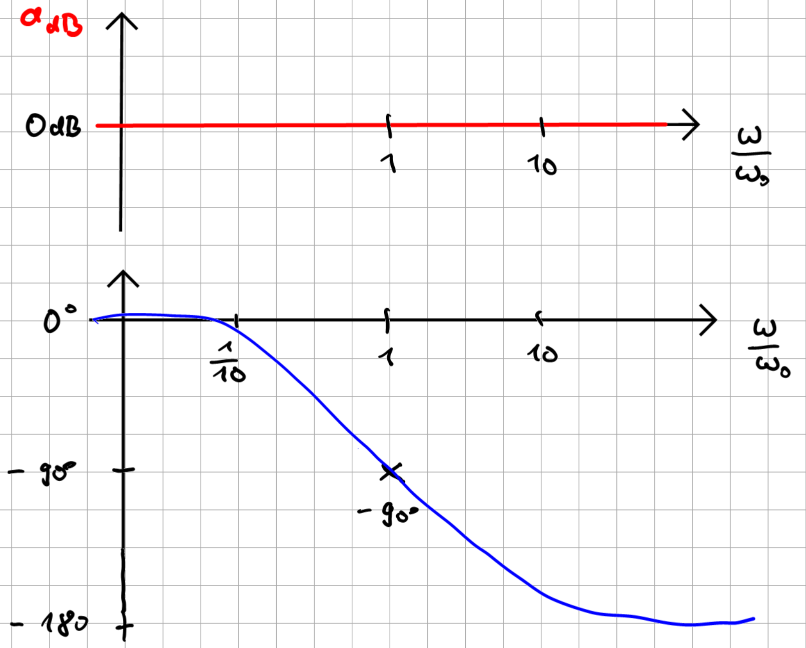
$$= 20 \log_{10} \sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2} - 20 \log_{10} \sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2}$$

$$= 0 \text{ dB} \; ! \qquad (\hat{=} \; 1)$$

$$\varphi_A = \varphi_Z - \varphi_N$$

$$= \sphericalangle(\underbrace{1-j\omega\tau}_{„Z2“}) - \sphericalangle(\underbrace{1+j\omega\tau}_{„Z1“})$$

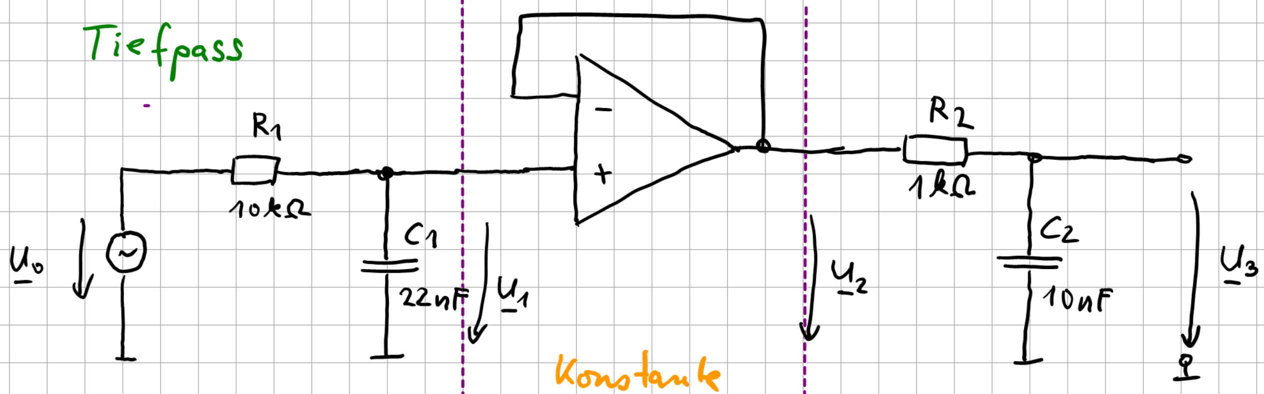
$\rightarrow -90°$ für $\omega \rightarrow \infty$ $\quad\quad$ $\rightarrow +90°$ für $\omega \rightarrow \infty$



Bsp: $\omega_0 = \frac{1}{s}$

9.1 Beispiel zum Bode-Diagramm

Tiefpass



$$\underline{A}_1 = \frac{\underline{U}_1}{\underline{U}_0} = \frac{\underline{Z}_{C1}}{R_1 + \underline{Z}_{C1}}$$

$$= \frac{\frac{1}{j\omega C_1}}{R_1 + \frac{1}{j\omega C_1}} \cdot \frac{j\omega C_1}{j\omega C_1}$$

$$\underline{A}_1 = \frac{1}{1 + j\omega \underbrace{C_1 R_1}_{T_1 = \frac{1}{\omega_1}}}$$

$$= \frac{1}{1 + j\frac{\omega}{\omega_1}}$$

$$\omega = 2\pi \cdot f$$

$$\omega_1 = 2 \cdot \pi \cdot f_1$$

$$= \frac{1}{1 + j\frac{\cancel{2\pi} f}{\cancel{2\pi} f_1}}$$

Tiefpass

$$\boxed{\underline{A} = \frac{1}{1 + j\frac{f}{f_1}}}$$

$$f_1 = \frac{\omega_1}{2\pi} = \frac{1}{2\pi T}$$

$$= \frac{1}{2\pi \; R_1 \cdot C_1}$$

$$= \frac{1}{2\pi \; 10k\Omega \cdot 22nF}$$

$$= \underline{\underline{723 \; Hz}}$$

$$\boxed{\underline{A}_2 = \frac{\underline{U}_2}{\underline{U}_1} = 1}$$